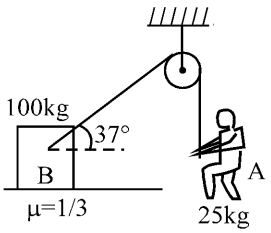
SECTION-I

Single Correct Answer Type **5 Q. [3 M (–1)]**

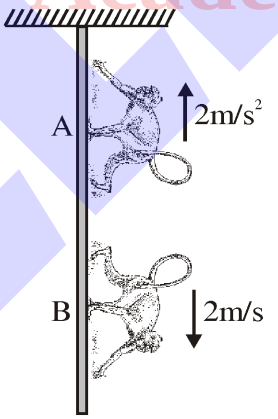
1. Block *B* of mass 100 kg rests on a rough surface of friction coefficient $\mu = 1/3$. A rope is tied to block *B* as shown in figure. The maximum acceleration with which boy *A* of 25 kg can climbs on rope without making block move is :



(A) $\frac{4g}{3}$ (B) $\frac{g}{3}$ (C) $\frac{g}{2}$ (D) $\frac{3g}{4}$

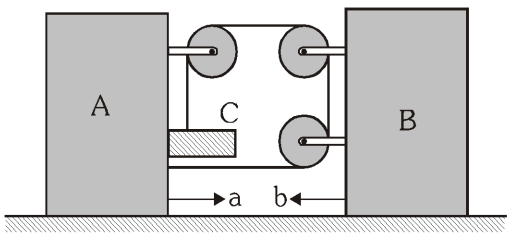
2. Two monkeys of masses 10 kg and 8 kg are moving along a vertical light rope, the former climbing up with an acceleration of 2 m/s^2, while the latter coming down with a uniform velocity of 2 m/s. Find tension in the rope at the fixed support.

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(A) 180 N (B) 200 N (C) 80 N (D) 216 N

3. If the blocks A and B are moving towards each other with acceleration a and b as shown in the figure. Find the net acceleration of block C.



(A) $a\hat{i} - 2(a+b)\hat{j}$ (B) $-(a+b)\hat{j}$ (C) $a\hat{i} - (a+b)\hat{j}$ (D) None of these

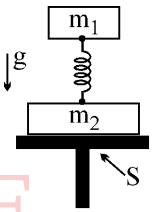
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4. The person in the drawing is standing on crutches. Assume that the force exerted on each crutch by the ground is directed along the crutch. If the coefficient of static friction between a crutch and the ground is 0.90, determine the largest angle θ_{max} that the crutch can have just before it begins to slip on the floor.



(A) $\tan^{-1}(0.9)$ (B) $\cot^{-1}(0.9)$ (C) $\sin^{-1}(0.9)$ (D) $\cos^{-1}(0.9)$

5. The system of two weights with masses m_1 and m_2 are connected with weightless spring as shown. The system is resting on the support S. The support S is quickly removed. The accelerations of each of the weights right after the support S is removed are.



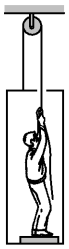
(A) $a_1 = 0, a_2 = \dfrac{(m_1+m_2)g}{m_2}$ (B) $a_1 = 0, a_2 = \dfrac{(m_1+m_2)g}{m_1}$

(C) $a_1 = \dfrac{(m_1+m_2)g}{m_1}, a_2 = 0$ (D) $a_1 = 0, a_2 = 0$

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Multiple Correct Answer Type 5 Q. [4 M (–1)]

6. A carpenter of mass 50 kg is standing on a weighing machine placed in a lift of mass 20 kg. A light string is attached to the lift. The string passes over a smooth pulley and the other end is held by the carpenter as shown. When carpenter keeps the lift moving upward with constant velocity :- ($g = 10\ m/s^2$)



(A) the reading of weighing machine is 15 kg.
(B) the man applies a force of 350 N on the string.
(C) net force on the man is 150 N.
(D) Net force on the weighing machine is 150 N.

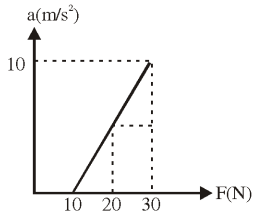
7. A block is kept on a rough surface and applied with a horizontal force as shown which is gradually increasing from zero. The coefficient of static and kinetic friction are $1/\sqrt{3}$ then



(A) When F is less than the limiting friction, angle made by net force on the block by the surface is less than 30° with vertical.
(B) When the block is just about to move, the angle made by net force by the surface on the block becomes equal to 30° with vertical.
(C) When the block starts to accelerate, the angle made by net force by the surface on the block becomes constant and equal to 30° vertical.
(D) The angle made by net force with vertical on the block by the surface, depends on the mass of the block.

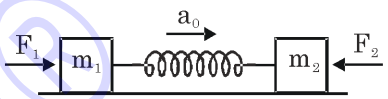
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8. A block placed on a rough horizontal surface is pushed with a force F acting horizontally on the block. The magnitude of F is increased and acceleration produced is plotted in the graph shown.



(A) Mass of the block is 2 kg.
(B) Coefficient of friction between block and surface is 0.5.
(C) Limiting friction between block and surface is 10 N.
(D) When $F = 8$ N, friction between block and surface is 10 N.

9. Two blocks m_1 and m_2 are connected with a compressed spring and placed on a smooth horizontal surface as shown in figure. Force constant of spring is k. Under the influence of forces F_1 and F_2, at an instant blocks move with common acceleration a_0. At that instant force F_2 is suddenly withdrawn. Mark **CORRECT** option.

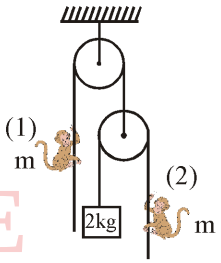


(A) Instantaneous acceleration of m_1 is $a_0 - \frac{F_1}{m_1}$

(B) Instantaneous acceleration of m_2 is $a_2 = a_0 + \frac{F_2}{m_2}$

(C) Instantaneous acceleration of m_1 is $a_1 = a_0$
(D) Spring force is $F_{spring} = m_2 a_0 + F_2$

10. Two monkeys (1) & (2) of same mass m = 1kg are hanging on the strings such that block of mass 2 kg remains at rest and it is given that monkey (2) is just holding the string. Then which of the following statement(s) are **CORRECT** ($g = 10$ m/s^2):-

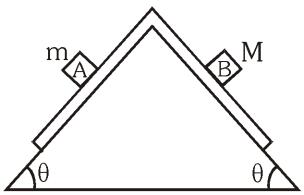


(A) Acceleration of monkey (2) is 10 m/s^2 upwards.
(B) Acceleration of monkey (1) is 30 m/s^2 upwards.
(C) Acceleration of monkey (1) with respect to his rope is 35 m/s^2 upwards.
(D) Acceleration of monkey (2) with respect to his rope is 50 m/s^2 upwards.

Linked Comprehension Type (2 Para × 3Q.) [3 M (-1)]
(Single Correct Answer Type)

Paragraph for Question no. 11 to 13

A light flat ribbon is placed over the top of the triangular prism as shown in figure. Two blocks are placed on the ribbon. The coefficient of static and kinetic friction between the ribbon and the blocks are μ_s and μ_k respectively. There is no friction between the ribon and the prism. Angle θ and the masses of the blocks m & M are given. Assume that M > m and blocks released simultaneously



11. If there is no slipping between ribbon and blocks then initial acceleration of the block will be

(A) $g\cos\theta\left[\frac{(M-m)}{(M+m)}\right]$ (B) $g\left[\frac{(M-m)}{(M+m)}\right]$ (C) $g\sin\theta\left[\frac{(M+m)}{(M-m)}\right]$ (D) $g\sin\theta\left[\frac{(M-m)}{(M+m)}\right]$

12. The tension in the ribbon when there is no slipping will be between ribbon and blocks

(A) $\mu_s mg\cos\theta$ (B) $\mu_s Mg\cos\theta$ (C) $\frac{2g\sin\theta mM}{M+m}$ (D) $\frac{g\sin\theta mM}{M+m}$

13. Now if we increase the angle θ then which block starts slipping first on the ribbon

(A) block A (B) block B (C) Simultaneously (D) Can't say anything

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Paragraph for Question No. 14 to 16

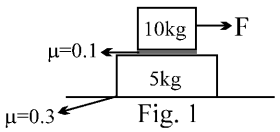


Fig. 1

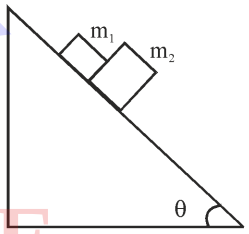
14. The maximum F which will not cause motion of any of the blocks is
(A) 10 N (B) 15 N (C) data insufficient (D) None

15. The maximum acceleration of 5 kg block is :-
(A) 1 m/s^2 (B) 3 m/s^2 (C) 0 (D) None

16. The acceleration of 10 kg block when $F = 30$ N is
(A) 2 m/s^2 (B) 3 m/s^2 (C) 1 m/s^2 (D) None

Matching List Type (4 × 4) 1 Q. [3 M (–1)]

17. A block of mass $m_1 = 1$ kg another mass $m_2 = 2$kg, are placed together (see figure) on an inclined plane with angle of inclination θ. Various values of θ are given in List I. The coefficient of friction between the block m_1 and the plane is always zero. The coefficient of static and dynamic friction between the block m_2 and the plane are equal to $\mu = 0.3$. In List II expressions for the friction on block m_2 are given. Match the correct expression of the friction in List II with the angles given in List I, and choose the correct option. The acceleration due to gravity is denoted by g.
[**useful information :** tan (5.5°) ≈ 0.1 ; tan(11.5°) ≈ 0.2 ; tan (16.5°) ≈ 0.3]



List–I		List–II	
(P)	θ = 5°	(1)	$m_2 g \sin\theta$
(Q)	θ = 10°	(2)	$(m_1 + m_2)g \sin\theta$
(R)	θ = 15°	(3)	$\mu m_2 g \cos\theta$
(S)	θ = 20°	(4)	$\mu(m_1 + m_2)g \cos\theta$

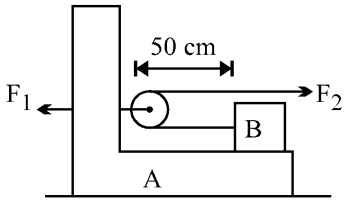
Code :
(A) P-1, Q-1, R-1, S-3 (B) P-2. Q-2, R-2, S-3
(C) P-2,. Q-2, R-2, S-4 (D) P-2, Q-2, R-3, S-3

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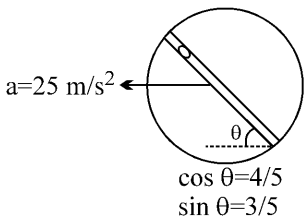
SECTION-II

Numerical Answer Type Question (upto second decimal place) 2 Q. [3(0)]

18. A 1kg block B rests as shown on a bracket A of same mass. Constant forces $F_1 = 20$N and $F_2 = 8$N start to act at time t = 0 when the distance of block B from pulley is 50cm. Time when block B reaches the pulley is _____.



19. A circular disc with a groove along its diameter is placed horizontally. A block of mass 1 kg is placed as shown. The co-efficient of friction between the block and all surfaces of groove in contact is $\mu = 2/5$. The disc has an acceleration of 25 m/s^2. Find the acceleration of the block with respect to disc.
[IIT-JEE 2006]

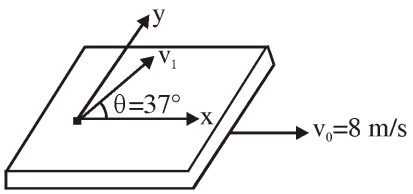


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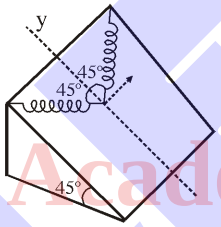
SECTION-III

Numerical Grid Type (Ranging from 0 to 9) 4 Q. [4 M (0)]

20. A board is being moved with a cosntant velocity v_0 on a smooth horizontal plane. A small block is projected on the rough boards. Coefficient of friction between board and block is $\mu=0.3$. Block is projected with velocity $v_1 = 25$ m/s at an angle of $\theta= 37°$ relative to ground. What is the velocity of block (in m/s) relative to ground after time t= 10 sec.

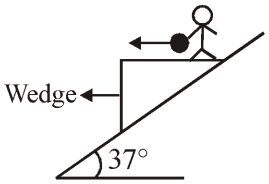


21. Figure shows a ball of mass m connected with two ideal springs of force constant k, kept in equilibrium on a smooth incline, suddenly right spring is cut. What is magnitude of instantaneous acceleration (in m/s^2) of ball.

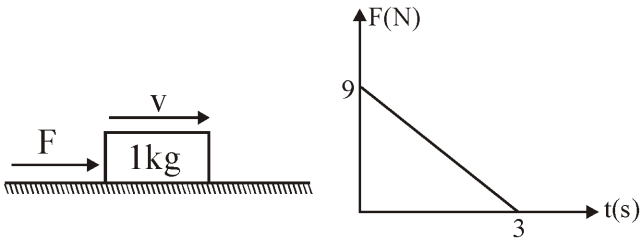


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22. A wedge can slide frictionlessly on a fixed incline of angle 37°. A girl who is strapped on the wedge [no relative motion between wedge and girl] pushes a small ball of negligible mass on the horizontal smooth upper surface of wedge as shown in figure. Ball is pushed at the same instant as the motion of wedge starts. Initial velocity of ball is 12 m/s relative to wedge. Find the time (in sec) after which ball returns to girl. Assume that horizontal surface of wedge is very long.



23. A 1 kg block moving on horizontal rough surface of friction coefficient 0.6 is pushed with a force varying with time t in seconds as shown in figure. If initial velocity of block was 4.5 m/s. Find the velocity (in m/s) at t = 3 s.

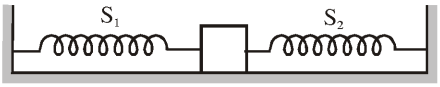


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SECTION-IV

Matrix Match Type (4 × 5) 1 Q. [8 M (for each entry +2(0)]

24. The figure shows a block B of mass 2 kg kept on a smooth horizontal floor in equilibrium with two identical springs of S_1 & S_2 force constant k = 100 N/m attached to it and to fixed supports as shown. The block is then displaced horizontally from this position by amount x and released. Match the initial equilibrium conditions & subsequent values of x in column-I with corresponding acceleration of the block when released in column II.

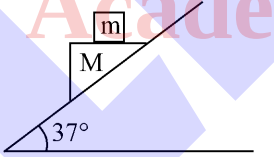


Column-I		Column-II	
(A)	S_1 and S_2 are relaxed and x = 2 cm.	(P)	6 m/s^2
(B)	S_1 and S_2 are stretched by 1 cm and x = 2 cm	(Q)	4 m/s^2
(C)	S_1 and S_2 are compressed by 2 cm and x = 4 cm	(R)	3 m/s^2
(D)	S_1 and S_2 are compressed by 4 cm and x = 1 cm	(S)	2 m/s^2
		(T)	1 m/s^2

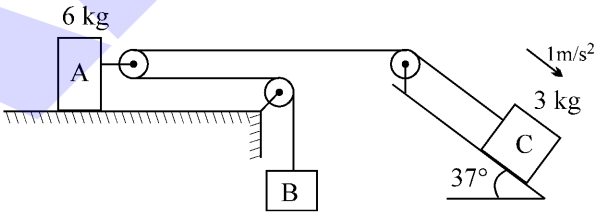
Subjective Type 6 Q. [4 M (0)]

25. Block *M* slides down on frictionless incline as shown. Find the minimum friction coefficient so that *m* does not slide with respect to *M*.

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26. Three blocks A, B & C are arranged as shown. Pulleys and strings are ideal. All surfaces are frictionless. If block C is observed moving down along the incline at 1 m/s^2. Find mass of block B, tension in string and accelerations of A, B as the system is released from rest.



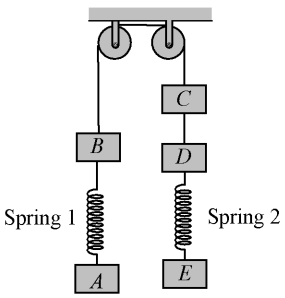
27. The system shown in the figure is initially in equilibrium. *A* is of mass 2*m* and *B*,*C*, *D* and *E* are of mass *m*. Certain actions are performed on the system. Every action has been taken individually when the system is intact. Find the direction and magnitude of acceleration of the blocks after each action of the following actions has been taken

(i) Spring 1 is cut

(ii) Spring 2 is cut

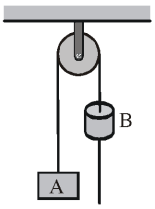
(iii) String between *C* and *D* is cut.

(iv) String between *B* and *C* is cut.

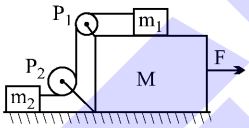


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28. A 2 kg block *A* is attached to one end of a light string that passes over an an ideal pulley and a 1 kg sleeve *B* slides down the other part of the string with an acceleration of 5 m/s^2 with respect to the string. Find the acceleration of the block, acceleration of sleeve and tension in the string. [g = 10 m/s^2]



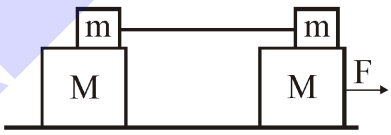
29. In the figure masses m_1, m_2 and M are 20 kg, 5 kg and 50 kg respectively. The co-efficient of friction between M and ground is zero. The co-efficient of friction between m_1 and M and that between m_2 and ground is 0.3. The pulleys and the string are massless . The string is perfectly horizontal between P_1 and m_1 and also between P_2 and m_2 . The string is perfectly vertical between P_1 and P_2.An external horizontal force F is applied to the mass M. Take g = 10 m/s^2.



(i) Draw a free-body diagram for mass M, clearly showing all the forces.

(ii) Let the magnitude of the force of friction between m_1 and M be f_1 and that between m_2 and ground be f_2. For a particular F it is found that $f_1 = 2 f_2$. Find f_1 and f_2 . Write down equations of motion of all the masses . Find F, tension in the string and accelerations of the masses.

30. Four blocks are arranged on a smooth horizontal surface as shown. The masses of the blocks are given (see the diagram). The coefficient of static friction between the top and the bottom blocks is μ_s. What is the maximum value of the horizontal force F, applied to one of the bottom blocks as shown, that makes all four blocks move with the same acceleration?



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GR # NLM & FRICTION — ANSWER KEY

SECTION-I

Single Correct Answer Type — 5 Q. [3 M (–1)]

1. (B) 2. (B) 3. (A) 4. (A) 5. (A)

Multiple Correct Answer Type — 5 Q. [4 M (–1)]

6. (A,B) 7. (A,B,C) 8. (A,B,C) 9. (B,C,D) 10. (A,B,C)

Linked Comprehension Type (Single Correct Answer Type) — (2 Para × 3Q.) [3 M (-1)]

11. (D) 12. (C) 13. (A) 14. (A) 15. (C)
16. (A)

Matching List Type (4 × 4) — 1 Q. [3 M (–1)]

17. (D)

SECTION-II

Numerical Answer Type Question (upto second decimal place) — 2 Q. [3(0)]

18. 0.5 19. 10

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) — 4 Q. [4 M (0)]

20. 8 21. 5 22. 5 23. 0

SECTION-IV

Matrix Match Type (4 × 5) — 1 Q. [8 M (for each entry +2(0)]

24. (A) - (S) ; (B) - (S) ; (C) - (Q) ; (D) - (T)

Subjective Type — 6 Q. [4 M (0)]

25. 3/4

26. $M_B = 15$ kg, $T = 15$, $a_A = 5$ m/s^2 , $a_B = 9$ m/s^2

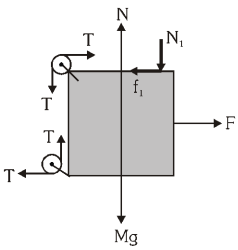
27. (i) $a_A = g\downarrow$, $a_B = \frac{2g}{3}\uparrow$, $a_c = \frac{2g}{3}\downarrow$, $a_D = \frac{2g}{3}\downarrow$, $a_E = 0$

(ii) $a_A = 0$, $a_B = \frac{g}{3}\downarrow$, $a_C = \frac{g}{3}\uparrow$, $a_D = \frac{g}{3}\uparrow$, $a_E = g\downarrow$

(iii) $a_A = 0, a_B = g\downarrow, a_C = g\uparrow, a_D = 2g\downarrow, a_E = 0$

(iv) $a_A = 0, a_B = 3g\downarrow, a_C = \frac{3g}{2}\downarrow, a_D = \frac{3g}{2}\downarrow, a_E = 0$

28. 5 m/s^2 downwards, 0 m/s^2, 10 N

29. (i) 

(ii) a = 3/5 m/s^2 , T = 18 N, F = 60N

30. $F = 2\mu_s mg\left(\frac{m+M}{2m+M}\right)$

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